

Solution Requirements

Date	26 March 2026
Project Name	Credit Card Approval Prediction System

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow applicants and bank officers to register and log in securely using email and password.	High
2	Authorization Levels	The system shall provide role-based access for applicants and bank officers to ensure secure access to features and prediction results.	High
3	External Interfaces	The system shall provide a user-friendly web interface for entering applicant details and displaying credit card approval predictions.	High
4	Transactions Processing	The system shall collect applicant information, validate inputs, process the data using the trained ML model, and generate instant approval or rejection predictions.	High
5	Reporting	The system shall maintain applicant records, prediction history, model performance, and approval results for future analysis and auditing.	Medium

6	Business Rules	Credit card approval shall be based on factors such as income, employment status, credit history, existing loans, payment behavior, and eligibility criteria evaluated by the trained machine learning model.	High
7	Compliance to Laws or Regulations	The system shall ensure the privacy and security of applicant data by following applicable data protection and banking security guidelines.	Medium
8	Other	The system shall support scalable deployment using Flask and IBM Watson Machine Learning for real-time credit card approval prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system shall process credit card applications and generate approval predictions with minimal delay..	Prediction results generated within 2 seconds .
2	Scalability	The system shall support multiple users and concurrent prediction requests without significant performance degradation.	Supports up to 1,000 concurrent users.
3	Security & Data Privacy	The system shall protect applicant information using secure authentication, encrypted data storage, and role-based access control.	Password encryption, secure authentication, and restricted access to sensitive data.
4	Reliability & Availability	The application shall remain operational with minimal downtime and provide accurate prediction results consistently.	99% system availability with reliable prediction processing.
5	Usability & Accessibility	The system shall provide a simple, responsive, and user-friendly interface that is accessible across different devices.	Users can complete a prediction in less than 3 minutes without assistance.

6	Other	The system shall be easy to maintain, update, and extend with new machine learning models or additional features.	Modular architecture with well-documented code and database structure.
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